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[24CSBSMA11]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY-2025

DISCRETE MATHEMATICS FOR COMPUTER SCIENCE

Time: 3 Hours **Maximum Marks: 100**

Instructions: i) Missing data may be suitably assumed

ii) Answer all Questions

		Questions	Blooms Level	CO's	Marks
		PART-A	**5 x2=10**		
1.	**a.**	Define Conjunction with the help of Suitable Example.	L1	CO1	**02**
	b.	Using Boolean algebra techniques, simplify $xyz + x\bar{y}z$.	L2	CO5	**02**
	c.	How many different strings of length 4 can be formed using the letters of the word FLOWER?	L2	CO2	**02**
	d.	Define Weighted graphs with suitable example.	L1	CO6	**02**
	e.	Let $A = \{1,2,3\}$ and $B = \{2,4,5\}$, determine the following. (i) $\lvert A \times B \rvert$ and (ii) Number of relations from A to B	L2	CO4	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Obtain the disjunctive normal form of the expression $[p \wedge \sim (q \wedge r)] \vee \sim (p \rightarrow q)$.	L2	CO1	**05**
		OR			
	b.	Determine the nature of the following proposition $(p \wedge (p \rightarrow q)) \rightarrow \sim q$.	L2	CO1	**05**
3.	**a.**	Construct truth table for the following Boolean expression $F(x, y, z) = xy + z$.	L3	CO5	**05**
		OR			
	b.	Express the Boolean function $F = xy + \bar{x}z$ as a product of maxterms.	L3	CO5	**05**
4.	**a.**	There are 50 baskets of apples, each basket containing no more than 20 apples. Show that there are at least 3 baskets containing the same number of apples.	L3	CO2	**05**
		OR			
	b.	Prove or disprove the statement that if x and y are real numbers then $[x^2 = y^2] \Leftrightarrow [x = y]$.	L3	CO2	**05**
5.	**a.**	Using generating functions find the non-negative integer solutions of the equation $x_1 + x_2 + x_3 + x_4 = 25$.	L3	CO3	**05**
		OR			

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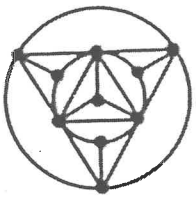
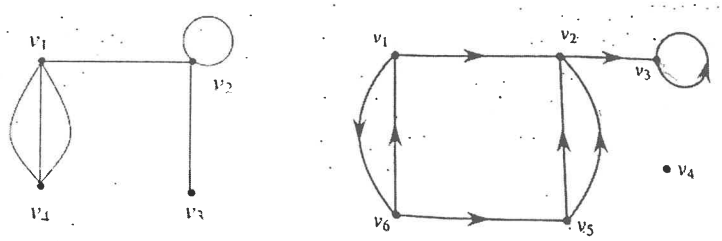
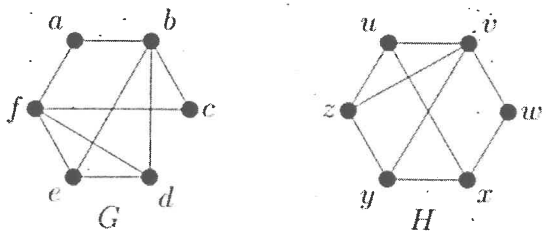
	b.	Solve the following recurrence relation $$3a_{n+1} - 4a_n = 0, \forall\, n \geq 0, a_1 = 5.$$	L3	CO3	**05**
6.	**a.**	Can there be a graph with 12 vertices such that two of the vertices have degree three each and the remaining 10 vertices have degree 4? If so find $\lvert E \rvert$.	L3	CO6	**05**
		OR			
	b.	Verify Euler's formula for the following graph 	L3	CO6	**05**
7.	**a.**	Let A and B be finite sets with $\lvert B \rvert = 3$. If there are 4096 relations from A to B then find $\lvert A \rvert$.	L2	CO4	**05**
		OR			
	b.	Show that the intersection of two subgroups of a group G is again a subgroup of G.	L2	CO4	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Obtain the Principle Disjunctive Normal Form and Principle Conjunctive Normal Form of $$p \vee [\sim p \rightarrow \{q \vee (\sim q \rightarrow r)\}].$$	L3	CO1	**10**
		OR			
	b.	With the following hypotheses: "It is not sunny this afternoon and it is colder than yesterday." "We will go swimming only if it is sunny." "If we do not go swimming, then we will take a canoe trip." "If we take a canoe trip, then we will be home by sunset." Using the inference rules, construct a valid argument for the conclusion: "We will be home by sunset."	L3	CO1	**10**
9.	**a.**	Sketch the logic circuits that produce the following outputs: (i) $(x+y+z)(\bar{x}\bar{y}\bar{z})$ and (ii) $xy + xz + yz$	L3	CO5	**10**
		OR			
	b.	Simplify the following SOP using Karnaugh Maps $$F = \sum (m_3, m_8, m_9, m_{10}, m_{11}, m_{12}, m_{13}, m_{14}, m_{15})$$	L3	CO5	**10**
10.	**a.**	In a class of 60 student appeared for an examination in which 25 students failed in Physics, 24 failed in Chemistry, 32 failed in Mathematics, 9 failed only in Physics, 6 failed only in Chemistry, 3	L3	CO2	**10**

		failed only in Physics and Chemistry, 5 failed only in Mathematics and Chemistry. Find how many failed in all three subjects and passed in all three subjects.			
		OR			
	b.	Use Mathematical Induction to prove the following $$1^2+2^2+3^2+\cdots\ldots\ldots\ldots\ldots+n^2=\frac{n(n+1)(2n+1)}{6}$$	L3	CO2	**10**
11.	**a.**	Determine the coefficient of x^{18} in the following functions (i) $(x+x^2+x^3+x^4+x^5)(x^2+x^3+x^4+\cdots..)^5$ (ii) $(x+x^3+x^5+x^7+x^9)(x^3+2x^4+3x^5+\cdots.)^3$	L3	CO3	**10**
		OR			
	b.	Solve the recurrence relation $D_n = bD_{n-1} - b^2D_{n-2}, \forall n \geq 3$ Given $D_1 = b > 0$ *and* $D_2 = 0$.	L3	CO3	**10**
12.	**a.**	Define vertex degree of an undirected graph, Indegree and outdegree of a directed graph. Find the vertex degree, indegree and outdegree of the following undirected and directed graphs 	L2	CO6	**10**
		OR			
	b.	Check whether the following graphs are isomorphic or not 	L3	CO6	**10**
13.	**a.**	Show that the following relation R defined on the set of all integers Z is an equivalence relation. $R = \{(x,y): x,y \in Z \text{ and } (x-y) \text{ is an even number}\}$.	L3	CO4	**10**
		OR			
	b.	Show that $G = \{1,2,3,4,5,6\}$ is an abelian group under multiplication modulo 7.	L3	CO4	**10**